Akhil Chigurupati

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PROFESSIONAL SUMMARY

Software engineer specializing in AI - with production experience building AI/ML-powered monitoring systems, real-time Kafka pipelines, and Python microservices across 500+ services at TCS. Independently built and shipped four projects spanning LLM-powered incident intelligence with RAG, LangChain, and RAGAS evaluation, distributed analytics, and NLP-based fake review detection published in Springer. Comfortable across the full stack - React frontends, FastAPI backends, Docker, Kubernetes, and CI/CD pipelines.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services | *Software Engineer* Apr 2023 - May 2024

- Developed Python microservices and REST APIs for an AI-driven infrastructure monitoring platform managing 500+ production services, reducing average debug cycle time by consolidating service boundaries and improving system modularity.
- Architected Apache Kafka streaming pipelines processing 50,000+ events/min for real-time log and metrics ingestion, reducing alert latency from minutes to seconds and enabling teams to detect incidents before user-facing impact.
- Built and integrated AI/ML anomaly detection models into production monitoring workflows covering 500+ services, flagging failure patterns 8-12 minutes before user-facing degradation was reported.
- Deployed containerized apps using Docker and Kubernetes managing 1,000+ pods with auto-scaling and high availability; automated incident remediation workflows, reducing manual resolution effort by 30-40% and improving MTTR.

PROJECT EXPERIENCE

AI Incident Intelligence Platform | *LLM-Powered Incident Analysis & Response*

Python · FastAPI · LangChain · LlamaIndex · OpenAI API · RAGAS · MLflow · Supabase pgvector · SSE · React · TypeScript · AWS (Lambda, SQS, DynamoDB, SNS) · Docker · GitHub Actions

- Architected an event-driven incident intelligence platform ingesting real-time metrics via AWS SQS, detecting anomalies with LLM-powered analysis, and streaming remediation recommendations to a TypeScript React frontend via SSE - LangChain retrieval chains and LlamaIndex index historical incidents in Supabase pgvector to ground every response.
- Built a RAGAS evaluation pipeline that automatically scores every RAG response for faithfulness, answer relevancy, and context precision; integrated MLflow to log eval scores and model configurations across iterations, enabling measurable, tracked improvement in response quality.

AI Log Analyzer | *AI-Powered Log Analysis & Anomaly Detection*

Python · FastAPI · Groq API · LLaMA 3 · sentence-transformers · Supabase pgvector · React · Vite · Docker Compose · GitHub Actions

- Built a full-stack log analysis tool where users upload server log files and query them in plain English - backend parses and chunks logs, embeds them using sentence-transformers, and stores vectors in Supabase pgvector for semantic search.
- Integrated Groq API (LLaMA 3) with a RAG pipeline to generate context-aware answers grounded in retrieved log chunks; built an automated anomaly detector that flags ERROR, WARN, and CRITICAL frequency spikes without LLM inference.

URL Shortener | *Distributed Analytics Platform*

Python · FastAPI · PostgreSQL · Redis · Apache Kafka · Prometheus · Grafana · Docker Compose · GitHub Actions · pytest

- Built a full-stack URL shortener with a real-time analytics pipeline using FastAPI, PostgreSQL, Redis, and Apache Kafka - decoupling click-event tracking from the redirect path via an async Kafka consumer worker.
- Implemented cache-aside pattern with Redis for sub-millisecond slug lookups; instrumented with Prometheus/Grafana; containerized with Docker Compose and automated CI/CD via GitHub Actions running pytest on every push.

Review Reality | *Fake Review Detection - Springer Published (Feb 2025)*

Python · Scikit-learn · Random Forest · Naive Bayes · MLP · Voting Classifier · NLP · Sentiment Analysis · Amazon/Yelp Dataset

- Developed a fake review detection system achieving 92% precision across 500,000+ review entries on 10 major platforms using ensemble ML algorithms (Random Forest, Naive Bayes, MLP, Voting Classifier) with advanced NLP feature engineering; reduced false positive rate by 30%.
- Implemented ensemble algorithms and software testing strategies to optimize prediction accuracy, resulting in a 15% increase in fraudulent review identification - validating model effectiveness at scale in peer-reviewed Springer publication.

EDUCATION

Regis University - Master of Science, Information Systems Denver, CO

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C, HTML, CSS
Backend: FastAPI, REST APIs, Microservices, API Development, System Design
Frontend: React.js, Vite, TypeScript, Server-Sent Events (SSE)
AI/ML & LLM Agents: OpenAI API, Claude API, Groq API, LLaMA 3, LangChain, LlamaIndex, RAG, RAGAS, MLflow, LLM Tool Calling, ReAct Agent Pattern, sentence-transformers, Scikit-learn, TensorFlow, Anomaly Detection
Cloud & DevOps: AWS (Lambda, SQS, DynamoDB, API Gateway, SNS, CloudWatch), Docker, Kubernetes, Helm, GitHub Actions, GitLab CI/CD, Linux, CI/CD Pipelines
Messaging & Streaming: Apache Kafka, AWS SQS
Databases: PostgreSQL, Redis, MongoDB, Supabase pgvector
Monitoring & Observability: Prometheus, Grafana, ELK Stack, MLflow, Infrastructure Monitoring
Core Competencies: Distributed Systems, Data Structures, Algorithms, Root Cause Analysis, Debugging, Incident Response